

Object-Oriented Programming in Java

Lecture 4 - Class Libraries

Emily Lucia Antosch

HAW Hamburg

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1. Introduction

1.1 Where Are We Currently?

- Last time we dealt with classes and objects.
- You can now
 - ▶ write simple classes in Java,
 - ▶ create objects from classes, use attributes and call methods and
 - ▶ use class variables and class methods.
- Today we continue with **Class Libraries**.

1.1 Where Are We Currently?

1. Introduction

1. Imperative Concepts
2. Classes and Objects
3. **Class Library**
4. Inheritance
5. Interfaces
6. Graphical User Interfaces
7. Exception Handling
8. Input and Output
9. Multithreading (Parallel Computing)

1.2 The Goal of This Chapter

- You apply strings, for example, for formatted output of data.
- You organize similar data in fields, matrices and lists.
- You convert strings to numerical values and apply mathematical functions to numerical values.

2. Strings

2.1 Strings

- Strings in C
 - ▶ Variables: Pointer to array of primitive data type char
 - ▶ Memory size managed by programmer
 - ▶ Data type has no methods
- Strings in Java:
 - ▶ Strings are objects of class String.
 - ▶ Variables reference objects
 - ▶ Memory size managed by object
 - ▶ Data type provides methods

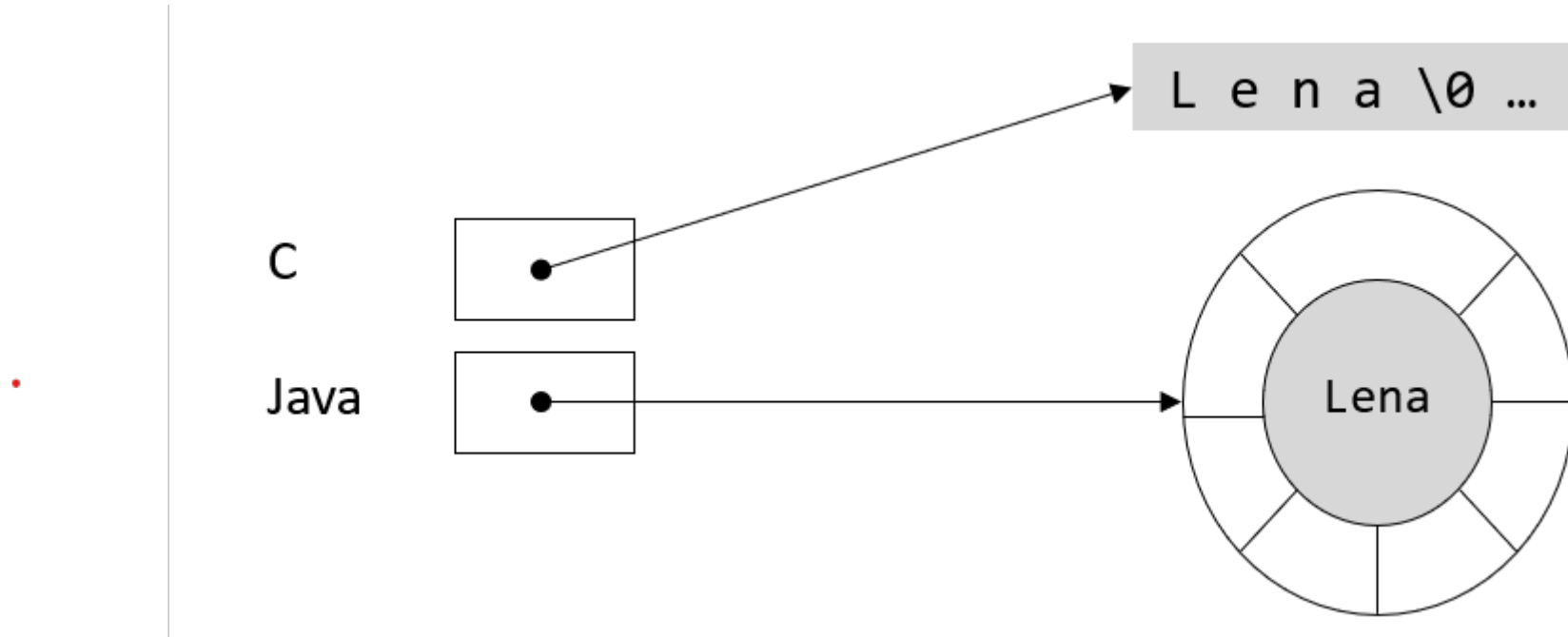


Figure 1: Strings: Differences between C and Java

2.2 Creating Strings

- Creation also using the `new` operator:

```
1 String name = new String("Lena");
```



- Alternatively by assigning a literal:

```
1 String name = "Lena";
```



- Assignment of a literal also possible after creation:

```
1 String name = new String("Lena");  
2 name = "Birgit";
```



2.3 Strings as Immutable

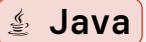
! Memorize

- As in many object-oriented languages:
 - ▶ Objects of class `String` are immutable.
 - ▶ Their value cannot be modified after creation.
 - ▶ For multi-stage construction, the `StringBuilder` class exists

? Question

What output does the following program produce?

```
1  public static void main(String[] args) {
2      String lena1 = new String("Lena");
3      String lena2 = lena1;
4
5      System.out.println("lena1: " + lena1 + "\nlena2: " + lena2);
6      System.out.println("References equal: " + (lena1 == lena2));
7
8      lena2 += " B.";
9      System.out.println("\nlena1: " + lena1 + "\nlena2: " + lena2);
10     System.out.println("References equal: " + (lena1 == lena2));
11 }
```



2.3 Strings as Immutable

- For illustration

```
1 String lena1 = new String("Lena");  
2 String lena2 = lena1;
```

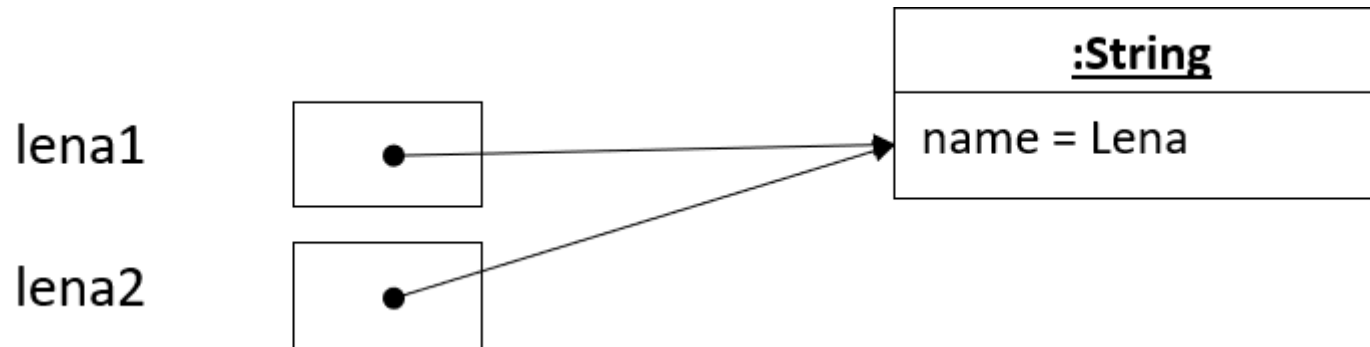
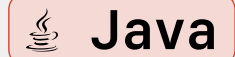


Figure 2: Reference to string

2.3 Strings as Immutable

2. Strings

```
1 lena2 += " B.";
```

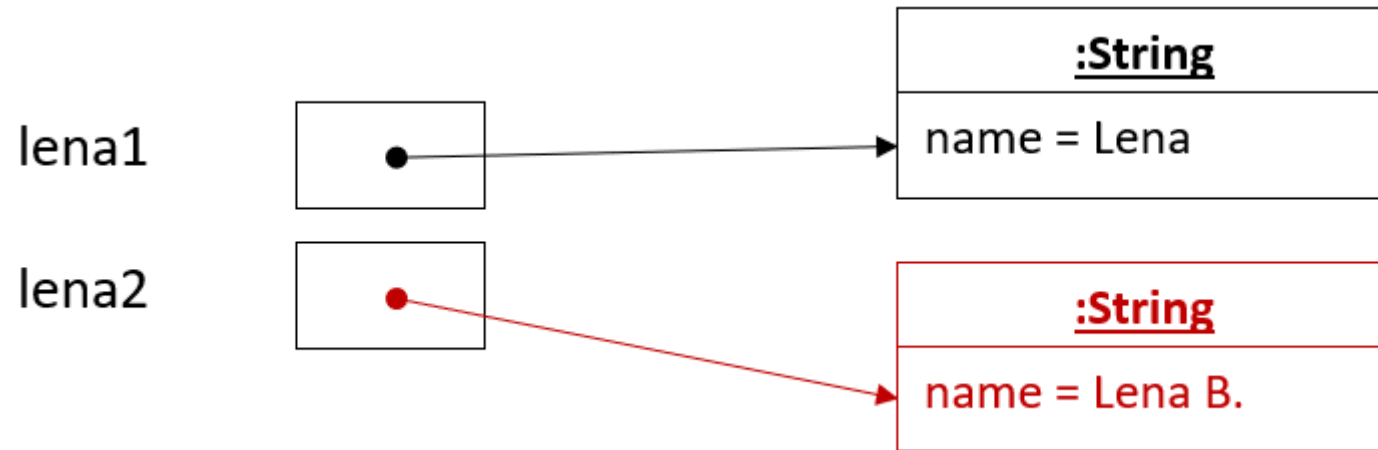
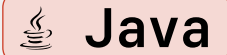


Figure 3: Changing the string leads to new object

2.4 String Concatenation

- Strings can be concatenated using the plus operator:

```
1 String name = "Lena " + "or " + "then ";  
2     name = name + "after all " + "again ";  
3     name += "Birgit?";
```



- Implicit conversion of other data types to a String object:
 - ▶ Evaluation of plus operators from left to right
 - ▶ Conversion to String if the other operand is not of type String

? Question

What will be output?

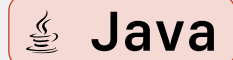
```
1 int a = 20;  
2 int b = 22;  
3 System.out.println("Year: " + a + b);  
4 System.out.println(a + b + " (Year)");
```



Java

2.5 toString() Method

```
1 public String toString() {  
2     // Method body  
3     // Return of an object of type String  
4 }
```



- Method header prescribed
- Method body freely programmable
- Returns a String for objects that should describe the object
- Is called during implicit conversion of an object to a String

Task 1

Let's try this out!

- Create a Person class and implement the toString() method
-
- Check the implicit call using console output.

2.5 toString() Method

```
1  public class Person {  
2      String firstName, surname;  
3  
4      public Person(String firstName, String surname) {  
5          this.firstName = firstName;  
6          this.surname = surname;  
7      }  
8  
9      public String toString() {  
10         return firstName + " " + surname;  
11     }  
12 }
```

2.5 toString() Method

```
1 Person lena = new Person("Lena", "Jensen");  
2 String name = "Name: " + lena;  
3 System.out.println(lena);  
4 System.out.println(name);
```

2.6 String Methods

- Additional methods include, for example:
 - ▶ Length of the string
 - ▶ Character at specific position (First character has index 0!)
 - ▶ Replace specific character
 - ▶ Search for specific character or substring
 - ▶ Split string
 - ▶ Conversion to lowercase or uppercase
 - ▶ Comparison of two strings
 - ▶ And several more!

Task 2

- Replace “Humbug” with “Hamburg”.

```
1 String hamburg = "Welcome to Humbug!";  
2 hamburg = hamburg.replace("Humbug", "Hamburg");  
3 System.out.println(hamburg);
```



? Question

What will be output?

```
1 String upper = "Welcome to Hamburg!";
2 String lower = "welcome to hamburg!";
3
4 System.out.println(lower.equals(upper));
5 System.out.println(lower.equals(upper.toLowerCase()));
6 System.out.println(lower.equalsIgnoreCase(upper));
```



2.7 String Formatting

- Often asked:
 - ▶ Can you also adjust the format of the string during output?
Yes, of course!
- Class method `format()`:
 - ▶ Creates a formatted string
 - ▶ No output to console occurs.
 - ▶ Syntax (almost) identical to `printf()` from C/C++

? Question

What will be output?

```
1  double wind = 21.4532;  
2  String weather = String.format("%s %d: %.1f km/h",  
    "Station", 7, wind);  
3  System.out.println(weather);
```



Java

? Question

What will be output?

```
1  double wind = 21.4532;  
2  String weather = String.format("%s %d: %.1f km/h",  
    "Station", 7, wind);  
3  System.out.println(weather);
```



Java

- **Output:** Station 7: 21.5 km/h

2.7 String Formatting

- Format specifications:

`%[ArgumentNo.][Flags][MinimumNumberCharacters][.Precision]`
Format

Format	Meaning
f, e, g	float
d	Integer
o	Octal
x, X	Hexadecimal
s	String
t	time
b	boolean

Flag	Meaning
-	Left Alignment
+	Print Prefix
0	Fill empty space with 0
,	Add commas for thousands
(	Add parentheses for negative numbers

Table 1: Formats and Flags

? Question

What will be the output?

```
1 double wind = 21.4532;  
2 System.out.println(String.format("%2.2f km/h", wind));  
3 System.out.println(String.format("%8.2f km/h", wind));  
4 System.out.println(String.format("%08.2f km/h", wind));
```

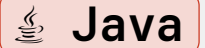


Java

? Question

What will be the output?

```
1 double wind = 21.4532;  
2 System.out.println(String.format("%2.2f km/h", wind));  
3 System.out.println(String.format("%8.2f km/h", wind));  
4 System.out.println(String.format("%08.2f km/h", wind));
```



- Output:
 - ▶ 21.45 km/h
 - ▶ 21.45 km/h
 - ▶ 00021.45 km/h

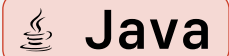
! Memorize

- Minimum number of characters:
 - ▶ Includes decimal places as well as the comma
 - ▶ Does not cut off any digits before the decimal point

? Question

- What do you notice?

```
1 double wind = 21.4532;  
2 System.out.println(String.format("%2.2f km/h", wind));
```

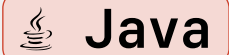


- Output: 21.45 km/h
-
-

? Question

- What do you notice?

```
1 double wind = 21.4532;  
2 System.out.println(String.format("%2.2f km/h", wind));
```

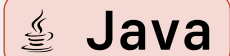


- Output: 21.45 km/h
- Above in output “German decimal comma” instead of “English point”
- Specified by localization



Example

```
1 double wind = 21.4532;
```



```
2 System.out.println(String.format(Locale.US, "%2.2f  
km/h", wind));
```

```
3 System.out.println(String.format(Locale.GERMAN,  
"%2.2f km/h", wind));
```

- Output: 21.45 km/h 21.45 km/h

3. Arrays

- Arrays in C:
 - ▶ Variables: Pointer to first element of the array in memory
 - ▶ Memory size managed by programmer
 - ▶ Data type has no methods
- Arrays in Java:
 - ▶ Arrays are objects.
 - ▶ Variables reference objects
 - ▶ Memory size managed by object
 - ▶ Data type provides methods

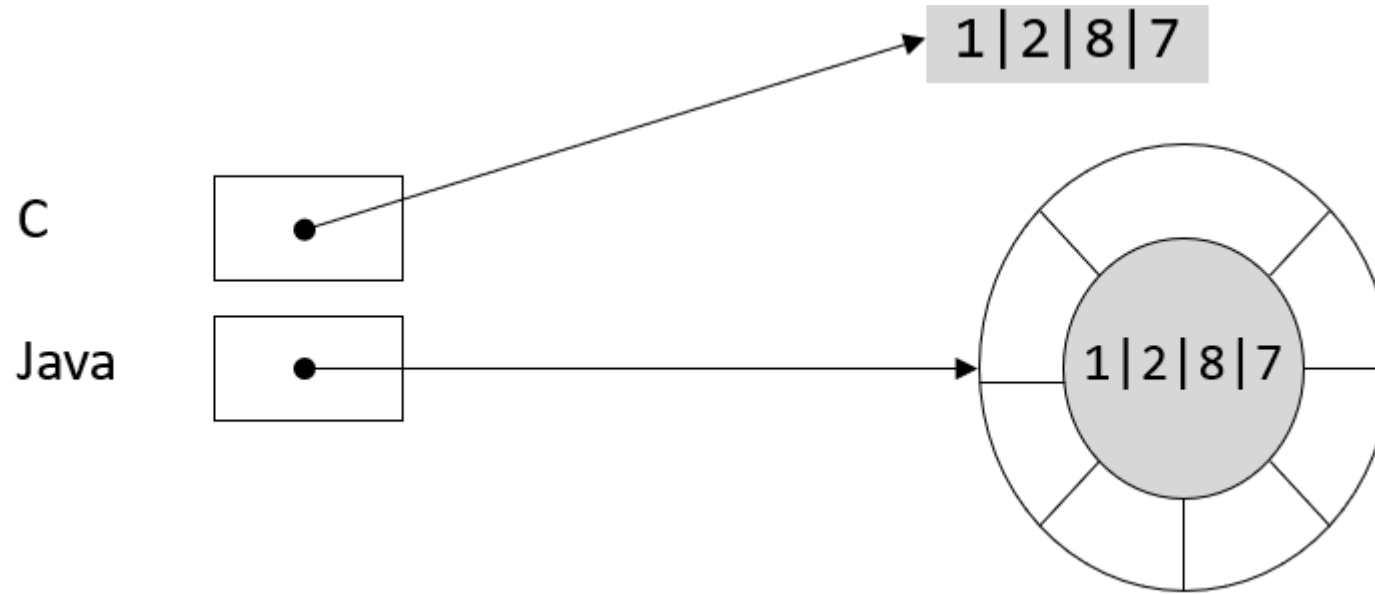


Figure 1: Arrays in Java and C

3.2 Creating Arrays

- Collection of elements with the same data type
- Data type becomes array through square brackets (e.g. `int[]`, `String[]`)
- Array classes are separate (additional) data types
- Declaration:
 - ▶ Does not require specification of length
 - ▶ Variable can reference arrays of any length
 - ▶ Declaration does not create object, but reference variable

3.2 Creating Arrays

3. Arrays

```
1 int[] filter;
```



Java

! Memorize

- Brackets after variable names allowed, but not recommended (Why?)

```
1 int filter[];
```



Java

3.3 Creation: Dynamic Declaration

- Create array object using new operator
- Number of fields in square brackets
- Note: No round “constructor brackets” after data type
- Values in array are initialized with 0, 0.0, false or null

```
1 int[] filter = new int[];
```

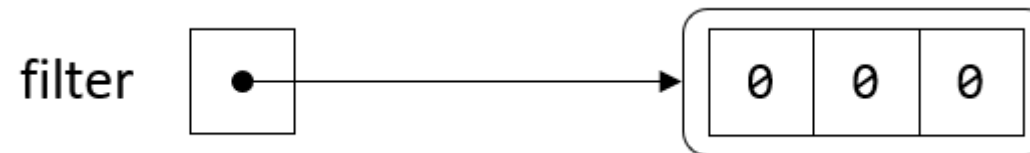
**Java**

Figure 2: Creation of an array

3.4 Creation: Assigning Elements

- Access to array element via index in square brackets
- First element has index 0

```
1 int[] filter = new int[3];  
2 filter[0] = 1;  
3 filter[1] = 2;  
4 filter[2] = 1;
```

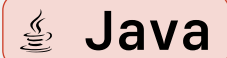


Figure 3: Assigning values through index access

3.5 Creation: Static Declaration

- You can assign values to an array already when creating the object.
- Values in curly braces and separated by commas
- Allowed with and without use of the new operator

```
1 int[] filter = {1, 2, 1};  
2 int[] filter = new int[] {1, 2, 1};
```

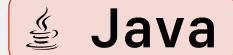


Figure 4: Filling during declaration

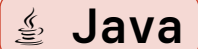
3.6 Properties: Array Classes

- Arrays are objects of the corresponding class:
 - ▶ Arrays have methods.
 - ▶ Number of elements via instance variable length

? Question

Which array is created by the code?

```
1 int[] filter = new int[3];
2 for (int i = 0; i < filter.length; i++) {
3     filter[i] = i * i;
4 }
```



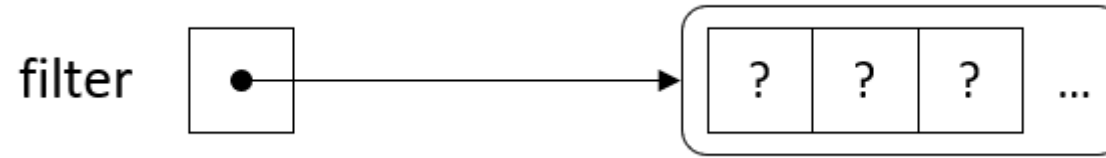


Figure 5: Filling through for loop

3.6 Properties: Array Classes

- Indices:
 - ▶ When accessing element, checks whether index is in allowed range
 - ▶ More in chapter on exceptions and error handling



Example

Examples of allowed and disallowed indices:

```
1 int[] filter = new int[3];
```

 **Java**

```
2 filter[0] = -1;
```

```
3 filter[2] = 4;
```

```
4 filter[-1] = 1;
```

```
5 filter[3] = 2;
```

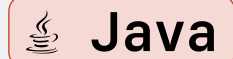


Figure 6: Indices of array `filter`

3.6 Properties: Array Classes

- Arrays can be declared for any data types (including custom classes)
- Objects must be of the same type (or subtype, more on this with inheritance)
- Not the objects stored, but references to the objects

```
1 Person[] friends = new Person[3];  
2 friends[0] = new Person("Lena");  
3 friends[1] = new Person("Birgit");  
4 friends[2] = new Person("Jan");
```

**Java**

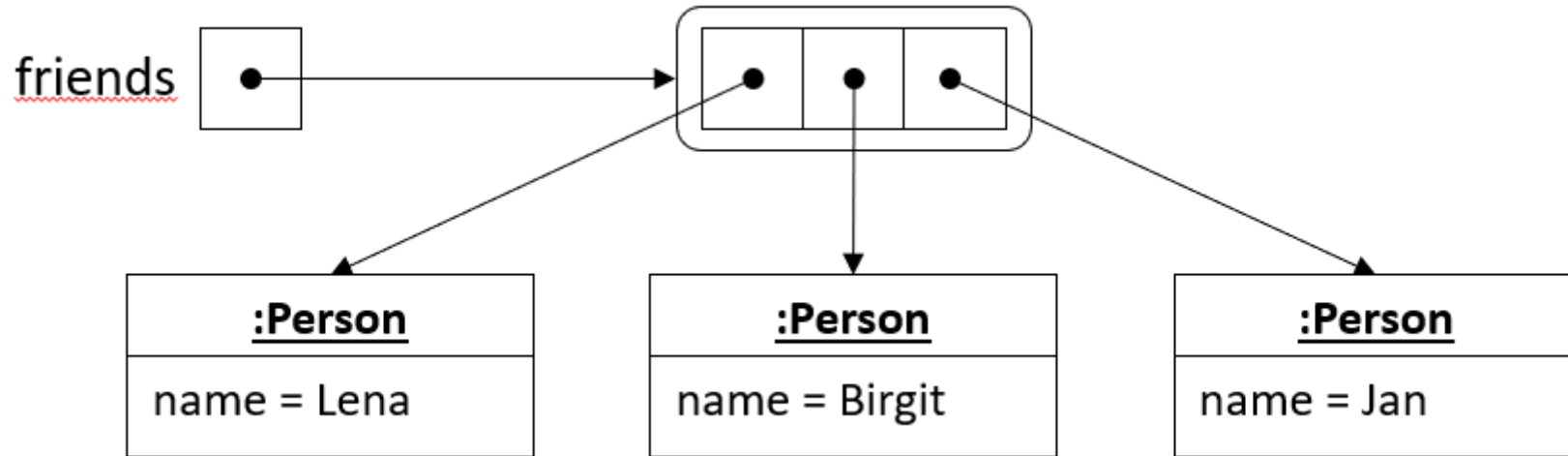


Figure 7: Arrays of objects

? Question

What will be output?

```
1 int[] a = {1, 2, 3, 4, 5};  
2 int[] b = a;  
3 b[3] = 0;  
4 System.out.println(b[2]);  
5 System.out.println(a[3]);
```



Java

? Question

What will be output?

```
1 int[] a = {1, 2, 3, 4, 5};  
2 int[] b = {1, 2, 3, 4, 5};  
3 System.out.println(a == b);
```



Java

? Question

What will be output?

```
1  public class ArrayDemo {
2      static int[] createSortedArray(int a, int b) {
3          if (a < b) {
4              return new int[] {a, b};
5          } else {
6              return new int[] {b, a};
7          }
8      }
9  }
```

 Java

3.7 Questions

3. Arrays

```
10      public static void main(String[] args) {  
11          System.out.println( createSortedArray(7, 4)[1] );  
12      }  
13  }
```

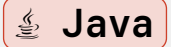
Task 3

- Write a method that sorts the elements of an `int[]` array in ascending order.
- Test the method using the array {10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 7}.

3.7 Questions

3. Arrays

```
1  public static void sort(int[] a) {
2      int i = 0;
3      while (i <= a.length - 2) {
4          if (a[i] > a[i+1]) {
5              // Swap elements and shift index to left element
6              int temp = a[i];
7              a[i] = a[i+1];
8              a[i+1] = temp;
9              if (i > 0)
10                 i--;
11          } else {
12              i++;
13          }
14      }
15 }
```



Java

4. Multidimensional Arrays

4.1 Multidimensional Arrays

4. Multidimensional Arrays

- Multidimensional arrays are “arrays of arrays”.
- Example: `int[][]` is array whose elements are of data type `int[]`.

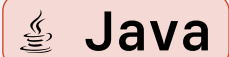
Dynamic declaration:

```
1  int[][] filter = new int[3][4];
```



Static declaration:

```
1  int[][] filter = {{1,2,3}, {4,5,6}, {7,8,9}};
```



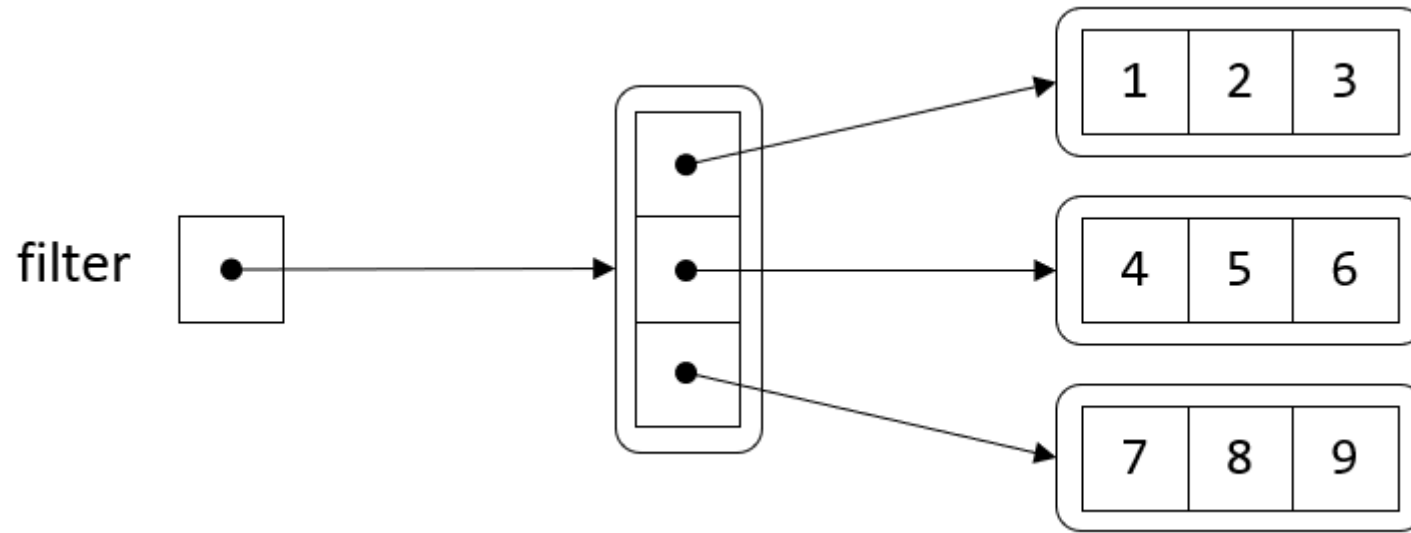


Figure 8: Multidimensional array

? Question

What will be output?

```
1  int[][] a = {{1,2}, {3,4}, {5,6}};
```

 **Java**

```
2
```

```
3  System.out.println(a.length);
```

```
4  System.out.println(a[2].length);
```

```
5
```

```
6  System.out.println(a[1][1]);
```

```
7  System.out.println(a[2][0]);
```

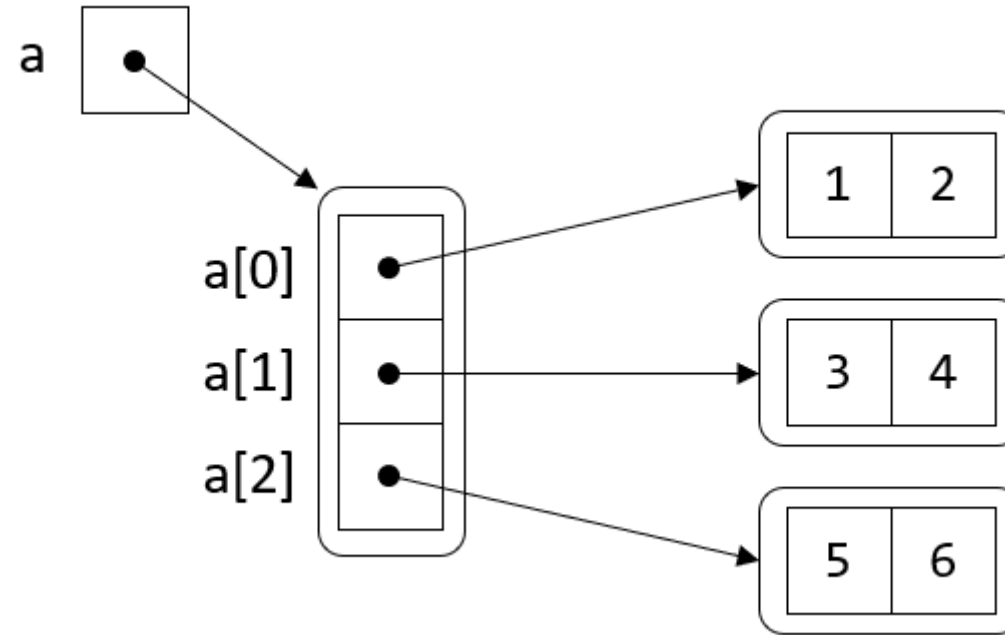


Figure 9: Multidimensional arrays with values

? Question

What will be output?

```
1  int[][] a = {{1,2}, {3,4}, {5,6}};  
2  int[] b = a[0];  
3  int c = b[1];  
4  
5  b[1] = 7;  
6  System.out.println(a[0][1]);  
7  System.out.println(c);
```



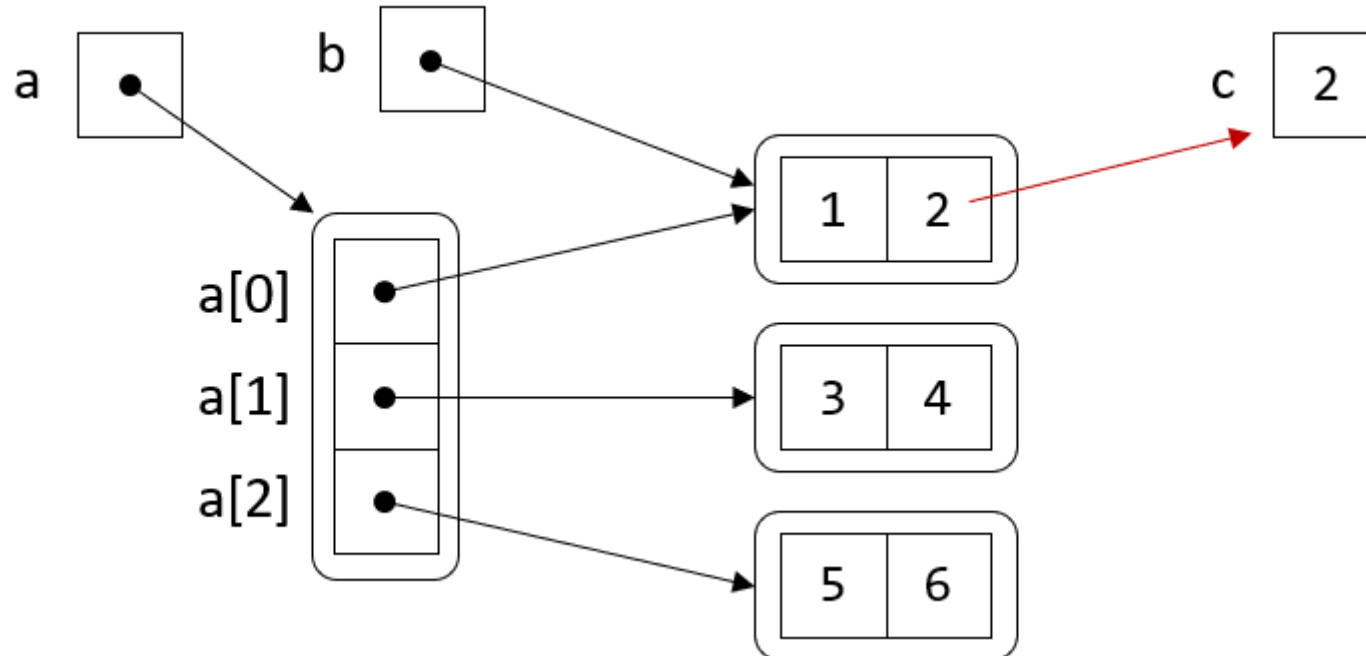


Figure 10: Complex multidimensional arrays

4.1 Multidimensional Arrays

4. Multidimensional Arrays

- Multidimensional arrays do not have to be rectangular
- Example: Assign own array to each row of a two-dimensional array

Task 4

Create a triangle matrix using a for loop!

4.1 Multidimensional Arrays

4. Multidimensional Arrays

- Multidimensional arrays do not have to be rectangular
- Example: Assign own array to each row of a two-dimensional array

Task 5

Create a triangle matrix using a for loop!

```
1  int[][] a = new int[3][];  
2  for (int i = 0; i < a.length; i++) {  
3      a[i] = new int[i + 1];  
4  }
```

 Java

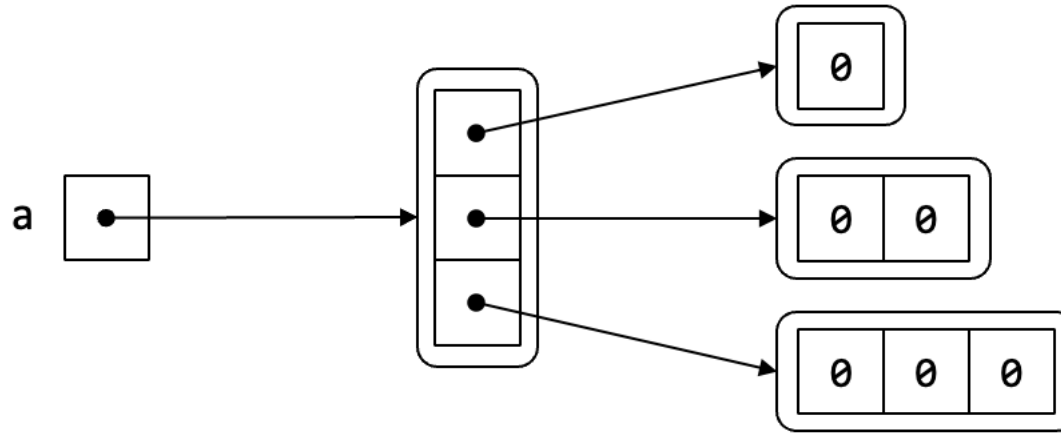


Figure 11: Multidimensional array in the shape of a triangle

5. Lists

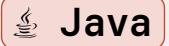
5.1 ArrayList

- Arrays: Size cannot be changed after creation (“semi-dynamic”)
- Lists: Elements can be added or removed (“dynamic”)
 - ▶ Data type of elements to be stored in angle brackets (see below: String)

5.1 ArrayList

5. Lists

```
1 public class ArrayListDemo {
2     public static void main(String[] args) {
3         ArrayList<String> names = new ArrayList<String>();
4         names.add("Lena");
5         names.add("Birgit");
6         names.add("Jan");
7         names.add(new String("Jan"));
8     }
9 }
```



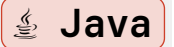
5.1 ArrayList

- Examples:
 - ▶ Number of elements (`size()`)
 - ▶ Access to elements (`get()`)
 - ▶ Query whether specific element is in list (`contains()`)
 - ▶ Remove element from list (`remove()`)

5.1 ArrayList

5. Lists

```
1  ArrayList<String> names = new ArrayList<String>();
2  String birgit = "Birgit";
3  names.add("Lena");
4  names.add(birgit);
5
6  for (int i = 0; i < names.size(); i++) {
7      System.out.println(names.get(i));
8  }
9
10 if (names.contains(birgit)) {
11     names.remove(birgit);
12 }
```



6. foreach Loop

6.1 foreach Loop

6. foreach Loop

```
1  for (DataType Variable : IterationObject) {  
2      Statements  
3  }
```



- Motivation:
 - ▶ Sometimes every element e.g. of an array or a list is needed
 - ▶ But: Position within the array or list is not needed
 - ▶ Therefore no loop counter as index needed

6.1 foreach Loop

- Loop iterates through array (or list) from first to last element:
- On first pass, variable has the value of the 1st element
- On second pass, variable has the value of the 2nd element and so on
- On last pass, variable has the value of the last element

? Question

What will be output?

```
1 int[] a = {7, 1, 3, 8};
```

 **Java**

```
2
```

```
3 for (int element : a) {
```

```
4     System.out.println("Element: " + element);
```

```
5 }
```

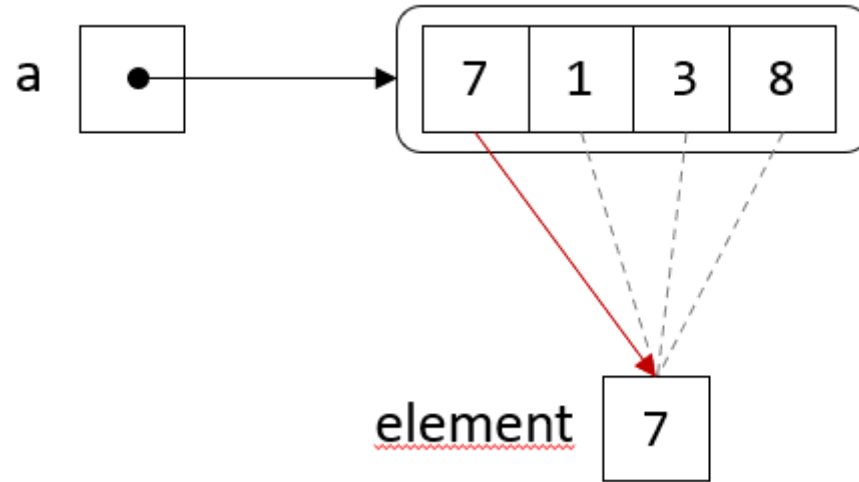


Figure 12: Result of foreach loop

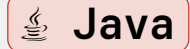
Task 6

- Create the following using a foreach loop:
- Method that returns the average of the numbers contained in an array
- Program that uses the method

6.1 foreach Loop

6. foreach Loop

```
1  static double average(double[] numbers) {
2      double sum = 0.0;
3
4      for(double number : numbers) {
5          sum += number;
6      }
7      return sum / numbers.length;
8  }
9
10 public static void main(String[] args) {
11     double[] a = {1.43, 2, .2, 6.32, 7.1, 8.1};
12     System.out.println("Average = " + average(a));
13 }
```



Java

7. Wrapper Classes & Math Class

7.1 Wrapper Classes

7. Wrapper Classes & Math Class

- Primitive data types:
 - ▶ Store value (e.g. integer) directly
 - ▶ Have no methods
- Wrapper classes:
 - ▶ “Wrap” primitive data types into classes
 - ▶ Provide methods (e.g. for integers)

7.1 Wrapper Classes

7. Wrapper Classes & Math Class

Primitive	Wrapper Class
boolean	Boolean
byte	Byte
short	Short
int	Integer
long	Long
char	Character
float	Float
double	Double

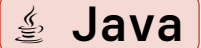
Table 1: Formats and Flags

7.1 Wrapper Classes

7. Wrapper Classes & Math Class

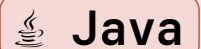
- Convert primitive data types to String

```
1  int a = 7;  
2  Integer b = new Integer(a);  
3  String c = b.toString();
```



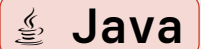
- Shorter alternative via class method:

```
1  String a = Integer.toString(7);
```



- Convert String to primitive data types:

```
1  String a = "7";  
2  int b = Integer.parseInt(a);
```

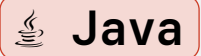


7.1 Wrapper Classes

7. Wrapper Classes & Math Class

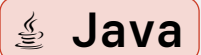
- Conversions:
 - ▶ Boxing: Conversion of primitive data type to object of wrapper class
 - ▶ Unboxing: Conversion of object of wrapper class to primitive data type

```
1 Integer object = new Integer(24); //Boxing of int value
2 int noObject = object.intValue(); //Unboxing of object
```



- Autoboxing: Automatic conversions (both directions)

```
1 Integer object = 24; //Automatic boxing of int value
2 int noObject = object; //Automatic unboxing of object
```



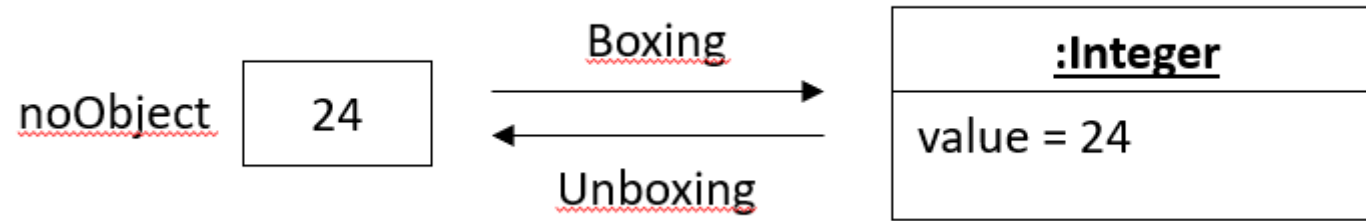


Figure 13: Type conversion with wrapper classes

- Mathematical constants: Euler's number e , pi π
- Mathematical functions (as class methods), e.g.:
 - ▶ Trigonometric functions
 - ▶ Rounding
 - ▶ Absolute value
 - ▶ Exponential function and logarithm
 - ▶ Maximum and minimum
 - ▶ Roots
 - ▶ Random numbers



Example

```
1 double angleDeg = 127.5;  
2 double angleRad = Math.toRadians(angleDeg);  
3 System.out.printf("cos(%.2f) = %.2f\n", angleRad,  
  Math.cos(angleRad));
```

**Java**

8. License Notice

8.1 Attribution

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- This work is based off of the work Prof. Dr. Marc Hensel.
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